Semih Yagli (**U.S. Lawful Permanent Resident**)

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GitHub: github.com/semihyagli ResearchGate: researchgate.net/profile/Semih-Yagli

Data Scientist with expertise in data analysis, machine learning, quantitative research and development, and full-stack application development, passionate about delivering creative, data-driven solutions to complex problems and generating actionable insights.

Educational Background

Princeton University, Princeton, New Jersey Sep. 2016 – May 2021
Electrical and Computer Engineering
Ph.D. in Electrical and Computer Engineering Dissertation Advisor: Prof. H. Vincent Poor
M.A. in Electrical Engineering GPA: 3.98/4.00

Orta Doğu Technical University, Ankara, Turkey (Double-Major) Sep. 2009 – May 2014
Mathematics, **Bachelor of Science**, GPA: 3.96/4.00
Electrical and Electronics Engineering, **Bachelor of Science**, GPA: 3.96/4.00

Patents

Machine Learning-Based Item Reranking Based on User Query and Cart Context	Patent No.: US20250342516A1	2025
Method and System for Fulfillment-Aware Product Search Ranking	(under review)	2026
Automatic Configuration Management in Cellular Networks	(under review)	2022

Coding and Technical

Programming Languages:	Python (XGBoost, Keras, PyTorch), Java, JavaScript, TypeScript, React, React Native, bash.
Data & Analytics:	SQL, NoSQL, BigQuery, R, Python (Pandas, PySpark, Matplotlib), Tableau.
Machine Learning & AI:	Neural Networks, Large Language Models, Natural Language Processing (NLP), Classification, Bayesian Inference, Thompson Sampling, Probabilistic Modeling, Clustering, Regression Analysis.
Development & Platforms:	Google Cloud (GCP), BigQuery, Firestore, Azure, Amazon Web Services, Axios HTTP.
Quantitative Skills:	Probability Theory, Optimization, Data Visualization.

Work Experience

Walmart Global Tech, Staff Data Scientist Oct. 2021 – Present

Lead end-to-end machine learning (ML) model development for e-commerce Search and BuyBox ranking systems, integrating engineering and data science capabilities across Python and Java. Conduct advanced experimentation, signal design, model training, and runtime deployment to drive large-scale business outcomes. Inform strategy through rigorous A/B testing, support infrastructure improvements, and mentor junior data scientists on navigating Walmart's complex systems.

Senior Data Analytics & Machine Learning Engineer at Search Ranking:

- **Delivered a 2.4% GMV lift** by architecting a unified ranking system that harmonized default and personalized search rankers, replacing fragmented legacy models with a high-performance, scalable framework.
- **Improved query-item relevance by 11%** by engineering a novel token-importance signal; spearheaded the end-to-end integration via **Java-based CI/CD** pipelines and advanced ML model deployments.
- **Directed cross-functional model validation** using interleaving and A/B testing frameworks, ensuring statistical rigor and driving consensus among senior stakeholders for high-stakes production changes.
- **Optimized system reliability and latency** by deploying high-throughput ranking features within core infrastructure.

Staff Data Scientist at Buy Box Offer Ranking:

- **Spearheaded Product Category feature development**, driving a site-wide **0.87% lift** in Multi-Offer Add-To-Cart (ATC) rates, generating an estimated **$58M in incremental annual revenue**.
- **Increased ATC conversions by 4%** by architecting a custom **Constrained Neural Network** to enforce price and shipping monotonicity, ensuring high-precision ranking logic.
- **Enhanced multi-offer conversion rates by 3.8%** by modeling multi-dimensional relationships between shipping speed, pricing, and buyer intent to optimize item-page rankings.
- **Designed a Bayesian explore–exploit framework** using **Thompson Sampling** to surface diverse resold (non-new) offer types on items with secondary offers, learning their conversion appeal under sparse feedback.

Technologies: Java (Runtime & Unit Testing), Python (PyTorch, Keras, Pandas, XGBoost), Google Cloud (GCP), BigQuery, Azure, Tableau, SQL.

Personal Projects and Experiences

Machine Learning Competitions May 2026 – Present

- **Published** "State-Driven Retrieval and Learned Re-Ranking for Conversational Music Recommendation" for the RecSys Challenge 2026, in *Proc. Workshop on the ACM RecSys Challenge*, pp. 82–86.
- **Competed** in the GLEE Competition (NeurIPS 2026 IAB Workshop), building AI agents for multi-turn bargaining, negotiation, and persuasion games against agents and humans; presenting a poster at the workshop.
- **Competing** in the Steerability Challenge (NeurIPS 2026), steering LLMs toward honesty without losing capability.

ACTUS / FINOS CDM Collaboration Sep. 2025 – Present

- **Leading** integration work extending the FINOS Common Domain Model (CDM) to natively support ACTUS cashflow calculations, bridging the gap between financial contract standards and open-source infrastructure.
- Driving design and implementation in collaboration with ACTUS-affiliated and independent researchers, building a reusable framework for standardized financial instrument modeling.

Smart Semi-Automated Investment & Portfolio Optimization System (On Hold) May 2025 – Present

- **Designed** a semi-automated investment system for portfolio optimization and long-horizon allocation decisions, with human-in-the-loop decision support driven by signals extracted from SEC EDGAR filings.
- **Built** a web data-acquisition pipeline to get SEC EDGAR filings, combining Selenium browser automation with HTTP extraction and BeautifulSoup parsing, plus data-quality validation.
- **Engineered** an Adaptive Kalman Filter-based allocation model, achieving a **4–6% gain** in information coefficient and an **8% reduction** in portfolio volatility across multiple market-regime backtests.

Technologies: Python (numpy, pandas, Selenium, BeautifulSoup, requests, yfinance, financetoolkit, financedatabase), Interactive Brokers (paper-trading account).

Advanced Pokémon Search – Creator & Sole Developer Jun. 2026 – Present

- **Built and shipped** a zero-backend, statically deployed search engine over **1,400+ species** across 30 game contexts, 350 competitive formats, and 27 regional Pokédexes, serving sub-second queries entirely client-side.
- **Designed** a downward-closed set-family algorithm resolving multi-constraint filter satisfiability in real time, replacing an intractable pairwise enumeration over 49K+ ability pairs per game context.
- **Extended** satisfiability to All/Any/None query modes with predictive feedback: each option shows, before it is picked, whether it would empty or conflict with the rest of the query, and the status line names the filters responsible.
- **Engineered** a TypeScript search core and a virtualized JavaScript front end, resynchronized weekly against upstream data with drift detection; guarded by **6,800+ automated tests** in four CI/CD pipelines, hardened by mutation testing.

Technologies: TypeScript, JavaScript, Node.js, esbuild, Mocha, jsdom, Playwright, Stryker, c8, ESLint, GitHub Actions.

Etiqueta – AI Data Labeling Platform, Founder & Sole Developer Mar. 2023 – May 2025

Architected and deployed **Etiqueta**, a first-of-its-kind mobile-native platform for AI-aided data labeling and image segmentation, built for gamified crowdsourcing across iOS, Android, and ARM-based desktop.

- **Curated 100K+ high-fidelity labels** and **13K image segmentations** via gamified workflows; published open-datasets on Kaggle and Hugging Face (1K+ downloads), plus a technical report and codebase on GitHub.
- **Launched on Global App Stores** as the only public, mobile-native data labeling solution, managing the full product lifecycle from UX design to store optimization.
- **Developed** a high-throughput React Native (web + native) interface supporting 30+ concurrent label interactions per screen, tuning DOM rendering and client-side caching to minimize interaction latency.
- **Built** a scalable Python (FastAPI) and Firestore backend exposing RESTful HTTP APIs with session-based OAuth, response caching, and unit testing (pytest) plus CI/CD for stability across iOS, Android, and macOS.

Technologies: Python (FastAPI), React Native, JavaScript, HTTP/REST, Firestore (NoSQL), OAuth, Axios, Pytest.

Awards and Honors

Excellence in Teaching Award at Princeton University 2018
Turkish Presidential Fellowship Award 2009 – 2014
Ranked 78th out of 1.5M+ candidates in the Turkish National University Entrance Exam 2008

Extra Curricular

Languages: Fluent in English, French, Spanish, and Turkish (Native).
Mentorship: Volunteer mentor at Princeton GradFutures program; each year mentoring about 4 students.
Leadership: Regional Princeton Graduate Alumni Leader; organizing networking events for alumni.

Ph.D. research highlights and full publication list available upon request.